NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

CCDS25029 - Can Deep Neural Networks Discover Chess Geometry Without Knowledge of Game Rules?

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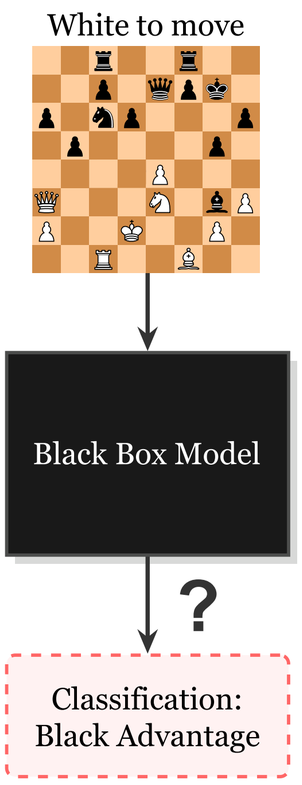
Introduction



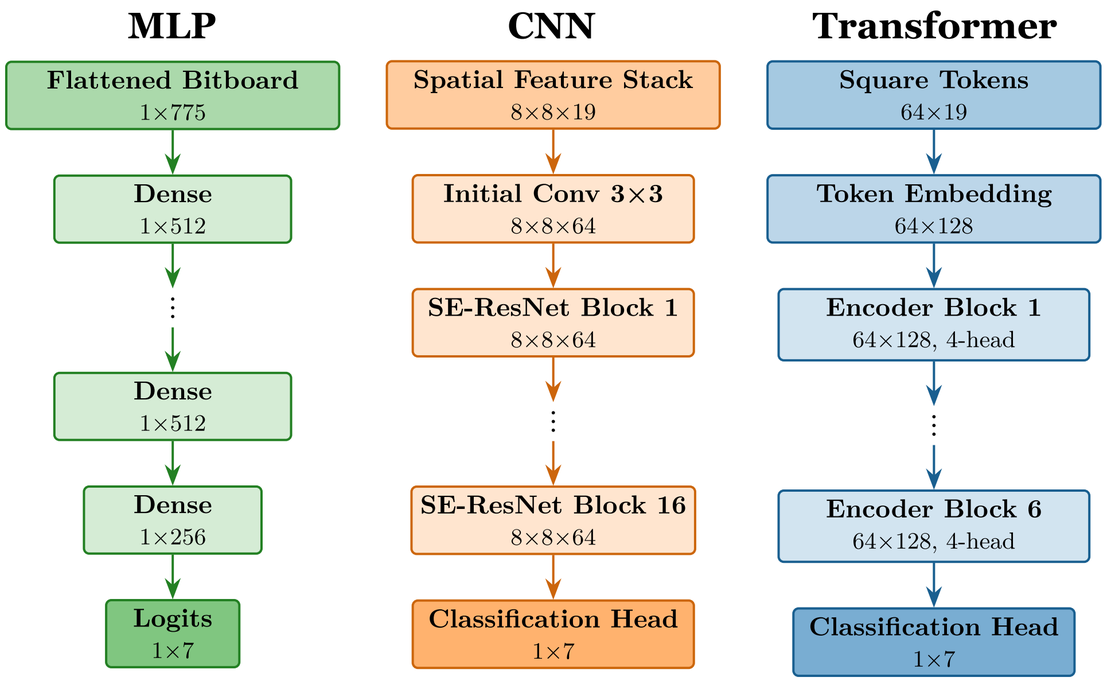
- Explainable AI is a crucial field of research, with chess often serving as a testbed with its perfect-information environment.
- Current frontier engines leverage various heuristics and deep search algorithms, obscuring the underlying contributions of their foundational architectures.
- We bypass these complexities to investigate the pure internal representations in the latent space formed solely by the networks.
- By visualizing and probing these internal representations, we reveal the different ways models learn the geometry of chess without rules, context, or search.

Methodology

In this study, we explore the differences in Multi-Layer Perceptrons (MLPs), Convolutional Neural Networks (CNNs) and Transformers via supervised learning. Models were trained to distill knowledge from Stockfish with only numerical encodings of positions as input.

Dataset

5.8M unique positions, augmented with random legal moves to introduce more imbalanced states, then class balanced for training.



Positional Accuracy (Evaluation Metric)

We coin this metric to measure model performance on states where material is perfectly equal, but the position evaluation is decisive.

Mechanistic Interpretability

Linear probes are trained on frozen intermediate activations of each model to predict legal piece movements.

Saliency Maps

Gradients are summed across all piece types (i) to determine which specific board squares (j) dictate the model's decision.

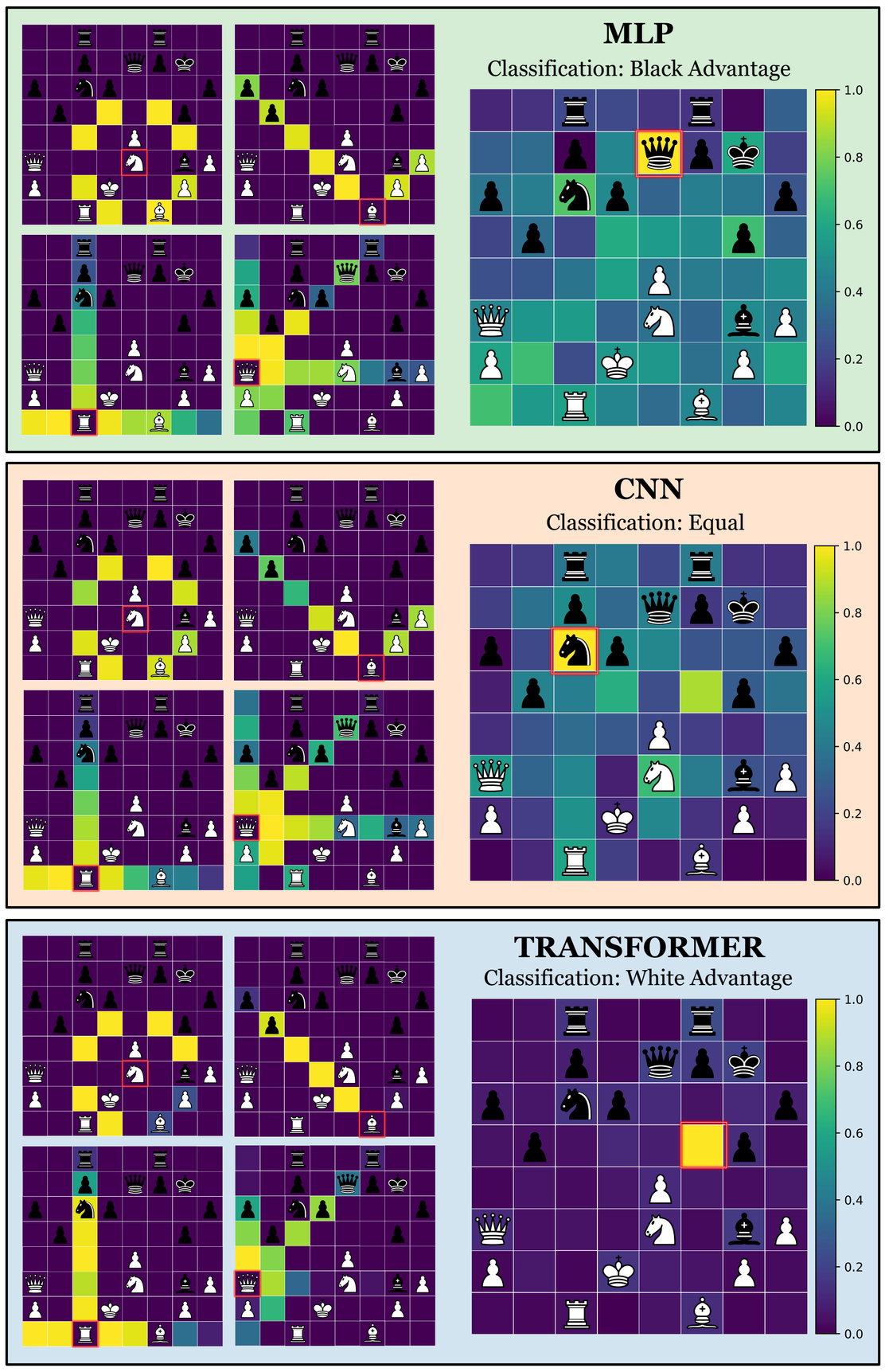
$$S_j = \sum_{i=1}^{12} \left| \frac{\partial y_c}{\partial x_{i,j}} \right|$$

Results

Performance comparison

Architecture	Params	Top-1 Acc.	Off-by-1 Acc.	MAE	Positional Acc.
MLP	1.32M	58.18%	88.05%	0.622	56.66%
CNN	1.22M	69.87%	94.72%	0.384	70.08%
Transformer	1.21M	**73.04%**	**95.83%**	**0.333**	**73.21%**

Legal Move Probe Maps **Saliency Maps**



The Transformer's self-attention mechanism captures global piece interactions far better than localized CNN convolutions or MLPs, which rely on raw capacity and default to simple piece counting.

Conclusion

We open these black boxes and proved that they can discover complex board geometry entirely from static states. We showed that emergent geometry depend heavily on inductive biases of the architectures when there is a lack of heuristics or search.